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Carbon- and energy-efficient ethanol electrosynthesis via interfacial cation enrichment

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1. Supplementary Methods

1.1 Multiphysics simulations

The simulation of cation distributions was performed in the COMSOL Multiphysics package based on the finite-element-method solver. The electric double layer at the electrode-electrolyte interface was modelled using the Gouy-Chapman-Stern (GCS) model, [1] which consists of the Stern layer, the diffuse layer, and diffusion layer (**Supplementary Figure 27**). The Stern layer consists of a monolayer of surface-adsorbed hydrated cations on the electrode surface, with a thickness of d_s depending on the hydrated cation, which is set to be 0.28 nm for H^+, 0.358 nm for Na^+, 0.33 nm for K^+, and 0.329 nm for Cs^+. The diffuse layer and diffusion layers contain free anions and cations distributed according to the Poisson and Boltzmann law, which forms the concentration gradient away from the electrode surface. In addition, the ICM and AEM were also considered in this work, specifically for pH simulations.

Generalised modified Poisson-Nernst-Plank (GMPNP)[2–4] with the homogeneous reaction term were applied to calculate the cations distribution, E-field, and pH in the diffuse and diffusion layer. The GMPNP equations are given by:

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot \boldsymbol{J_i} + \sum_p R_i \qquad \textbf{(Equation S1)}$$

$$\boldsymbol{J_i} = D_i \nabla C_i - \frac{D_i C_i z_i F}{RT} \nabla \phi - \frac{D_i C_i N_A a_i^3 \sum_i \nabla C_i}{1 - \sum_i a_i^3 C_i} \qquad \textbf{(Equation S2)}$$

$$\nabla \cdot (\varepsilon_0 \varepsilon_e \nabla \phi) = -\rho_f = -F \sum_i z_i C_i \qquad \textbf{(Equation S3)}$$

In Equation S1, C_i is the concentration of the ith species, $\boldsymbol{J_i}$ is the flux for the ith species, and R_i is the rate of production of the ith species due to the homogeneous reaction as given by equation (S9)-(S13). In Equation S2, D_i is the diffusion constant of the ith species, a_i is the

solvation size of the ith species, z_i is the valence charge of the ith species, F is the Faradaic constant, R is the ideal gas constant, T is the temperature, N_A is the Avogadro constant, and ϕ is the space potential in the electrolyte. In Equation S3, ε_0 is the vacuum permittivity, and ε_e is the relative permittivity of the electrolyte solution.

As defined in Gouy-Chapman-Stern (GCS) model,[5] the OHP is the closet proximity of the ions to the electrode surface. Therefore, the Stern layer contains no charged species but water. Thus, the charge density at OHP, σ_{OHP}, is given by:[3]

$$\sigma_{OHP} = \varepsilon_0 \varepsilon_e \left.\frac{d\phi}{dx}\right|_{x_{OHP}} = \frac{\varepsilon_0 \varepsilon_{Stern}(\phi_M - \phi_{OHP})}{d_{Stern}} \qquad \textbf{(Equation S4)}$$

where ε_{Stern} and d_{Stern} denote the relative permittivity in and thickness of the Stern layer, respectively, ϕ_M and ϕ_{OHP} are the potentials at the electrode surface and OHP, respectively. Equation S4 specifies the relationship between the potential (ϕ) and the potential gradient ($\frac{d\phi}{dx}$) at the OHP.

To account for the E-field dependence of the dielectric permittivity in the diffuse and diffusion layers, the Booth model, equation (S5), is employed. [6–8]

$$\varepsilon_e(E;\beta) = n^2 + (\varepsilon_r(0) - n^2)\frac{3}{\beta E}\left[\coth(\beta E) - \frac{1}{\beta E}\right] \qquad \textbf{(Equation S5)}$$

In equation S5, $E = |-\nabla\phi(r)|$ is the local E-field strength, n is the electrolyte refractive index, and β is a parameter given by: [9]

$$\beta = \frac{5\mu}{2kT}(n^2 + 2) \qquad \textbf{(Equation S6)}$$

where μ is the dipole moment of the water molecule.

Furthermore, the value of ε_{Stern} was extracted from Equation S7 and S8 as suggested by Johnson et al.[3] to consider the effect of the electric field on dielectric permittivity in the stern layer.

$$C_{Stern} = \frac{\varepsilon_0 \varepsilon_{Stern}}{x_{Stern}} \qquad \textbf{(Equation S7)}$$

$$C_{Stern} = -1.8923\, V^3 - 4.4698\, V^2 + 2.0741V + 26.323 \qquad \textbf{(Equation S8)}$$

Even though Equation S8 being derived from experiments in Na electrolyte, it provides a more plausible estimation of ε_{Stern} when juxtaposed with the scenario of presuming ε_{Stern} equals ε_e.

The diffusion rate of each species within the ion-exchange layer is reduced compared to that in the solution. Given its porous structure, the effective diffusivity should be accounted for within the porous domain during simulation, which depends on the porosity and tortuosity according to Bruggeman relationship.[10] More importantly, the fixed charges within the ion-exchange layer introduces a decelerating effect on the diffusion rate of the counter ionic species. Nevertheless, the precise quantitative assessment of the porosity, tortuosity, and fixed-charge density in the ion-exchange layer poses significant challenges. Consequently, we opted for a simplified modeling approach, presuming a tenfold reduction in the diffusion rate of counter ions within the ion-exchange layer. This choice provides a qualitative exploration of the impact of ion-exchange layer thickness on local pH variations using GMPNP model.

In the simulations, the reactions S9-S13 are considered for the homogeneous reactions occurring within the electrolyte solution.

$$CO_2 + H_2O \xleftrightarrow{k_{1f},\ k_{1r}} HCO_3^- + H^+ \quad (K_1) \qquad \textbf{(Equation S9)}$$

$$HCO_3^- \xleftrightarrow{k_{2f},\ k_{2r}} CO_3^{2-} + H^+ \quad (K_2) \qquad \textbf{(Equation S10)}$$

$$CO_2 + OH^- \overset{k_{3f},\ k_{3r}}{\longleftrightarrow} HCO_3^- \quad (K_3)$$ **(Equation S11)**

$$HCO_3^- + OH^- \overset{k_{4f},\ k_{4r}}{\longleftrightarrow} CO_3^{2-} + H_2O \quad (K_4)$$ **(Equation S12)**

$$H_2O \overset{k_{wf},\ k_{wr}}{\longleftrightarrow} H^+ + OH^- \quad (K_w)$$ **(Equation S13)**

The heterogeneous reactions S14 and S15 are considered as the electrochemical reactions occurring on the planar electrode surface during the CO_2RR process:

$$CO_2(aq) + H_2O + 2e^- \rightleftarrows \mathrm{CO(g)} + 2OH^-$$ **(Equation S14)**

$$2H_2O + 2e^- \rightleftarrows H_2(\mathrm{g}) + 2OH^-$$ **(Equation S15)**

The flux of the solution species at the surface is then given by equations (S16)-(S18):

$$\vec{J}_i\big|_{x=0,t} = 0 \quad i = HCO_3^-, CO_3^{2-}, K^+, H^+, Na^+, Cs^+$$ (**Equation S16**)

$$\vec{J}_{CO_2}\Big|_{x=0,t} = \frac{1}{n_{e,CO}} \times \frac{j_{tot}}{F} \times FE_{CO}$$ (**Equation S17**)

$$\vec{J}_{OH^-}\big|_{x=0,t} = \frac{1}{n_{e,OH^-}} \times \frac{j_{tot}}{F}$$ (**Equation S18**)

where F is the Faraday's constant, t is the time, $n_{e,CO}$ is the number of electrons consumed to reduce CO_2 to CO and n_{e,OH^-} is the number of electrons consumed per OH^-.

The rate of each species R_i, can be broken into carbonate equilibria as follows:

$$R_{CO_2} = \left(-[CO_2]_{aq}\,[H_2O]k_{1f} + [H^+][HCO3^-]\,k_{1r}\right) + \left(-[CO_2]_{aq}\,[OH^-]k_{3f} + [HCO3^-]\,k_{3r}\right)$$ **(Equation S19)**

$$R_{CO_3{}^{2-}} = \left([HCO_3{}^-]\,k_{2f} - [H^+]\left[CO_3{}^{2-}\right]k_{2r}\right) + \left([HCO_3{}^-]\,[OH^-]k_{4f} - [H_2O]\left[CO_3{}^{2-}\right]k_{4r}\right)$$ **(Equation S20)**

$$R_{HCO_3{}^-} = \left([CO_2]_{aq}\,[H_2O]k_{1f} - [H^+][HCO_3{}^-]\,k_{1r}\right) + \left(-[HCO_3{}^-]\,k_{2f} + [H^+]\left[CO_3{}^{2-}\right]k_{2r}\right) + \left([CO_2]_{aq}\,[OH^-]k_{3f} - [HCO_3{}^-]\,k_{3r}\right) + \left(-[HCO_3{}^-]\,[OH^-]k_{4f} + [H_2O]\left[CO_3{}^{2-}\right]k_{4r}\right)$$ **(Equation S21)**

$$R_{H^+} = \left([CO_2]_{aq}\,[H_2O]k_{1f} - [H^+][HCO_3{}^-]\,k_{1r}\right) + \left([HCO_3{}^-]\,k_{2f} - [H^+]\left[CO_3{}^{2-}\right]k_{2r}\right) + \left([H_2O]k_{wf} - [H^+][OH^-]\,k_{wr}\right)$$ **(Equation S22)**

$$R_{OH^-} = \left(-[CO_2]_{aq}\,[OH^-]k_{3f} + [HCO_3{}^-]\,k_{3r}\right) + \left(-[HCO_3{}^-]\,[OH^-]k_{4f} + [H_2O]\left[CO_3{}^{2-}\right]k_{4r}\right) + \left([H_2O]k_{wf} - [H^+][OH^-]\,k_{wr}\right)$$ **(Equation S23)**

The values of the rate constants for reactions S9-S13 are given in **Supplementary Table 5**. The gas concentration of CO_2 is assumed to obey Henry's law as given by

$$C^0_{CO_2,aq} = K^0_H\, C_{CO_2,g}$$ (**Equation S24**)

Where K^0_H is Henry's constant and is given as a function of temperature T by equation (S25):

$$\ln K^0_H = 93.2417 * \left(\frac{100}{T}\right) - 60.2409 + 23.3585 * \ln\left(\frac{T}{100}\right)$$ (**Equation S25**)

where T is the temperature and is assumed to be 298.15 K in this study. The saturated concentration of CO_2 in an electrolyte is then given by equation (S26):

$$\log\left(\frac{C^0_{CO_2,aq}}{C_{CO_2,aq}}\right) = K_S C_S$$ (**Equation S26**)

In equation (S26), C_S is the molar concentration of the electrolyte and K_S is the Sechenov's constant and can be calculated by the Equation S27:

$$K_S = \sum(h_{CO_2} + h_{ion})$$ (**Equation S27**)

$$h_{CO_2} = h^0_{CO_2} + h^T_{CO_2}(T - 298.15)$$ (**Equation S28**)

Values of h for all species can be found in **Supplementary Table 6**.

The intricacy of the EDL is heightened by factors such as Faradic currents, interfacial chemical interactions, and phenomena at molecular scales, where quantum mechanics becomes more pertinent than continuum theory. Additionally, the distinct behavior of hydrated ions under

a strong electric field, as compared to in a dilute bulk solution, also complicates the study of the EDL, especially with adsorption. Existing models like MPB, GMPNP, and SMPNP, while well-established, do not completely encapsulate the complexity of the EDL. Nonetheless, these models remain useful tools for visualization and modeling of the EDL. The GMPNP and SMPNP models offer more reliable and flexible approaches for modeling the EDL compared to the MPB model. In light of this, the GMPNP model was implemented to describe the EDL in this paper, reflecting a more accurate representation of EDL phenomena.

1.2 Density functional theory (DFT) calculations. DFT was applied to calculate the Gibbs free energy change of various HCCO* protonation processes, including HCCO* to HCCOH*, HCCO* to HCCHO*, HCCO* to H_2CCO*, on Cu and Cu-Ag surfaces. The structure of HCCO* + H* was also calculated to study the effect of adsorbed H* on the HCCO* protonation processes. All the DFT calculations were carried out in the Vienna ab initio simulation package[11–13] with a plane wave pseudo-potential implementation[14]. The exchange-correlation function was described by the spin-polarized generalized gradient approximation of Perdew-Burker-Ernzerhof[14] and the electron-ion interactions was described by projector augmented wave potentials[15,16]. The kinetic cut-off energy of 450 eV was used for the plane-wave expansion. The $6 \times 3 \times 1$ and $12 \times 6 \times 1$ Monkhorst-Pack[17] k-point meshes were used for geometry optimization and electronic structure calculation. The long-range Van der Waals interactions were described by the zero-damping DFT-D3 method of Grimme et al.[18]

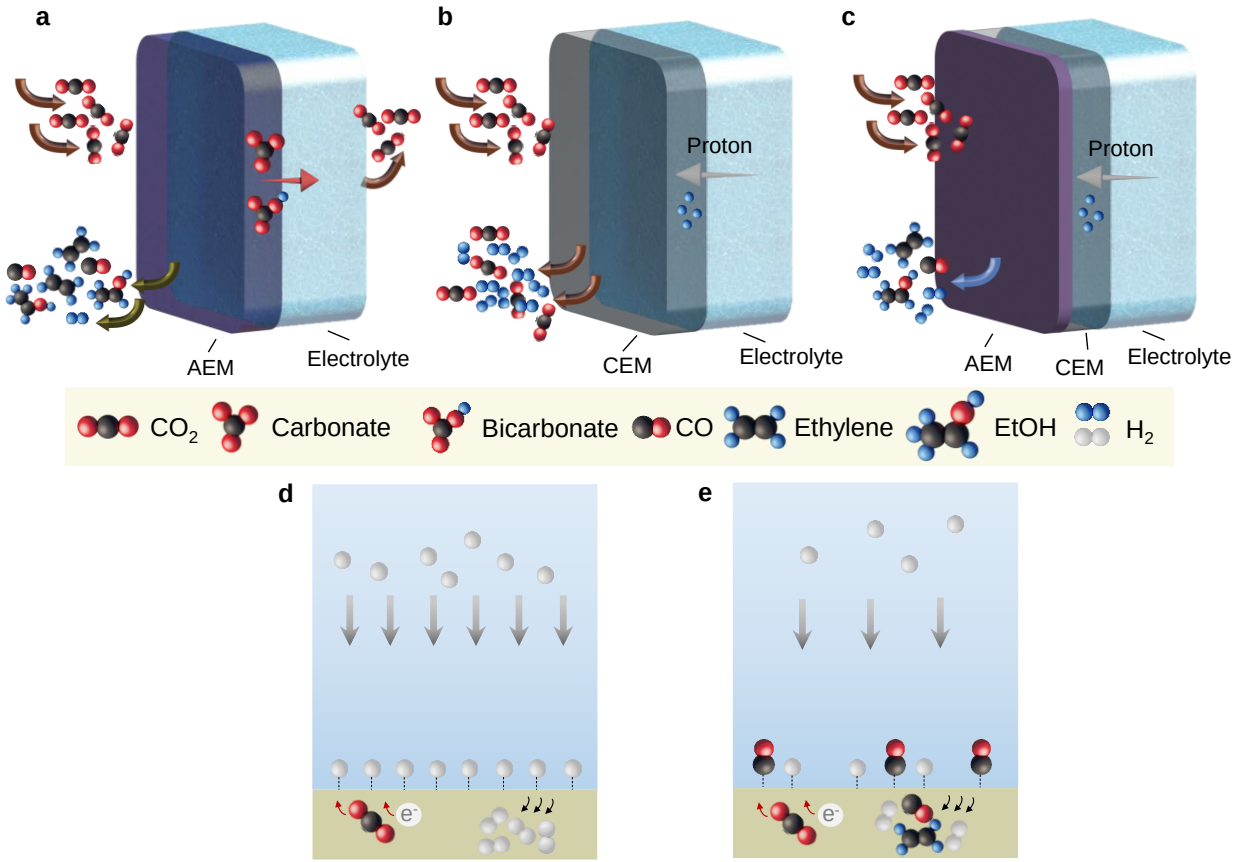
The (111) surface of face-centred cubic Cu was considered here using a $6 \times 3 \times 4$ periodic cell, since the (111) surface is found to be the dominant surface in the Cu-based electrocatalyst material utilized in this work, which is proved by the XRD measurements. 15 Å of vacuum space was set among the periodic surfaces in the z-direction to decouple the interaction between them. Ag-Cu surface was constructed by placing Ag clusters, including a 3-atom and 9-atom cluster, on the Cu(111) surface. Ag-doped Cu surface was constructed by replacing Cu atoms with Ag atoms. To reduce the computational cost, all atoms in the two bottom-most layers were fixed during the structural optimization calculations while other metal atoms and the adsorbates were allowed to relax.

The computational hydrogen electrode model[19] was used to calculate the Gibbs free energy by correcting the electronic energies directly determined from DFT calculations with Zero-point

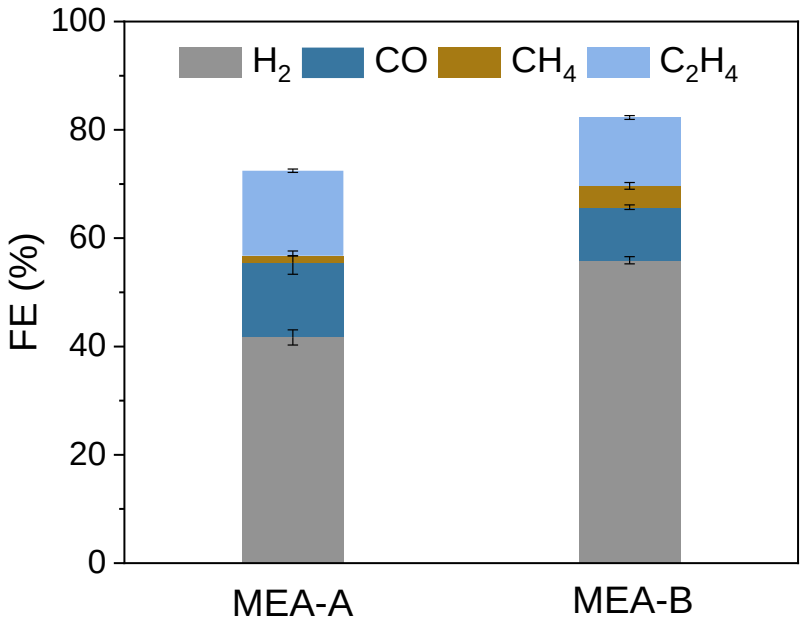
energies, entropies, and heat capacities, which are calculated from harmonic oscillator approximation at 298.15 K. Contributions to Gibbs free energies for each gas species and adsorbates are summarized in **Supplementary Table 7**. We note that the Gibbs free energies determined from our calculations provide a reasonable prediction of semiquantitative thermodynamic trends electrochemical conditions, since we ignore the presence of transition states and charged intermediates on surfaces.

Charge-density difference[20,21] was applied to investigate the charge transfer between the HCCOH* intermediate and surface. To consider the field and solvation effects, a hexagonal charged water overlayer-that is, five water molecules and one hydronium (H_3O^+)[22] – was introduced. To optimize the structure of the charged water overlayer, Ab initio molecular dynamics simulations were conducted in canonical ensemble (NVT) with the Nose-Hoover thermostat[23–25] and a 1.0 fs time step at 300 K, as performed in our previous study.[26] Reaction intermediates were included in the optimized geometry from AIMD simulations, and again to perform DFT calculations.

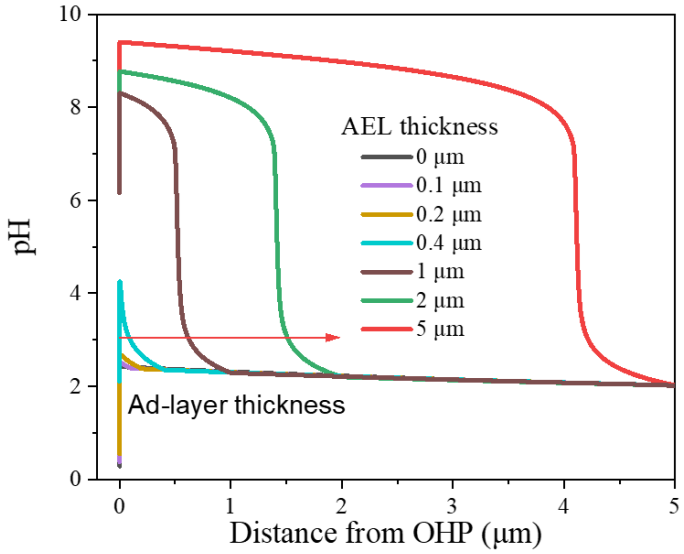
Supplementary Note 1 | Discussion on different MEA electrolyzer design and local microenvironment. MEA CO_2 electrolyzers with a cation exchange membrane (CEM) combined with acidic electrolyte were introduced to mitigate the CO_2 cross over issue (**Supplementary Figure 1**). In this configuration, the (bi)carbonate ion crossover would decrease, if not eliminated, due to the CEM-enabled acidity.[27] However, the unshielded cathode surface substantially favors the hydrogen evolution reaction (HER) over CO_2R due to the high availability of protons and the fast kinetics of the HER (**Supplementary Figure 1d**).[27,28] Using a bipolar membrane in a forward-bias bipolar membrane (f-BPM) mode (**Supplementary Figure 1c**) in a MEA enables high carbon efficiency and CO_2R.[27,29] In this configuration, the AEM would maintain a local alkaline environment and enable (bi)carbonate ion transport towards the CEM, while the CEM provides protons. Due to carbonate equilibria, the (bi)carbonate ions react with protons to regenerate CO_2 and H_2O between the AEM and CEM, which tremendously decreases the CO_2 loss to the anode.[27,28] By depositing an AEM ad-layer on the catalyst surface, the local pH would increase and favor CO_2R (**Supplementary Figure 1e**).[27]



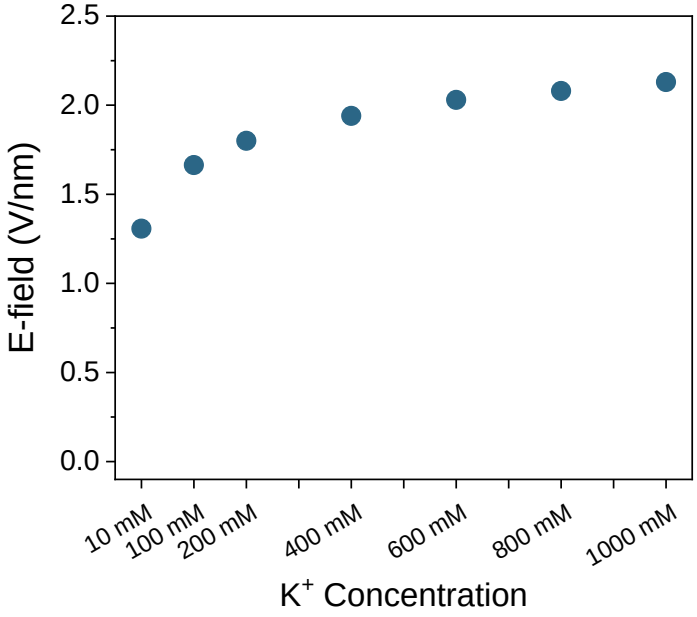
Supplementary Figure 1. Different configurations of CO_2 MEA electrolyzers. (a) MEA with an AEM membrane which results in >70% CO_2 crossover to anode via (bi)carbonate. (b) MEA with a CEM membrane that minimizes CO_2 crossover (< 30%). (c) MEA with a forward bias-BPM that reduces the CO_2 crossover with a basic local cathode surface. (d) Microenvironment in an MEA with a CEM membrane. The catalyst surface is expected to be too acidic due to proton accumulation at the outer Helmholtz plane (OHP). Due to the fast kinetic nature of HER, these accumulated protons can be easily reduced to H_2 by the electron on the electrode. (e) Microenvironment in a f-BPM MEA which favors CO_2RR by providing local alkalinity.



Supplementary Figure 2. CO_2R performance of Cu catalyst coated with benzimidazolium (an anion exchange ad-layer) in MEA. CO_2R performance at different anolyte concentration in MEA: (MEA-A) 0.01 M H_2SO_4 and (MEA-B) 0.2 M H_2SO_4. CO_2R was performed at 200 mA cm^{-2} for these experiments. The anion exchange layer coating is 3 mg cm^2. The plotted values are means of three replicates, while error bars indicate standard deviation.

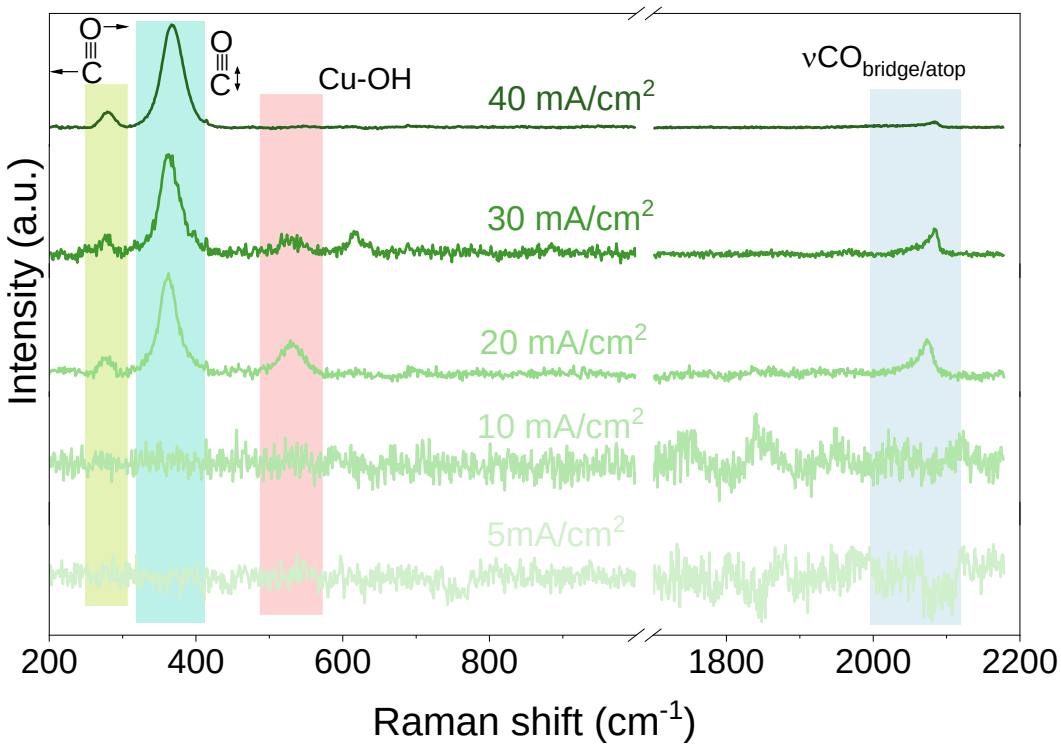


Supplementary Figure 3. Modelling results of local pH in the presence of anion exchange layer. Modeled pH changes away from the outer Helmholtz plane (OHP) with different anion exchange layer thicknesses under an applied current density of 100 mA cm^{-2}. If no anion exchange layer was used, the catalyst surface is expected to be too acidic due to proton accumulation at the outer OHP (as schematically shown in Supplementary Figure 1d). Due to the fast kinetic nature of HER, these accumulated protons can be easily reduced to H_2 by the electron on the electrode. However, by depositing an anion exchange layer on the catalyst surface, the local pH would increase. More alkalinity is expected as a thicker anion exchange layer is deposited which favors CO_2R (Supplementary Figure 1e).

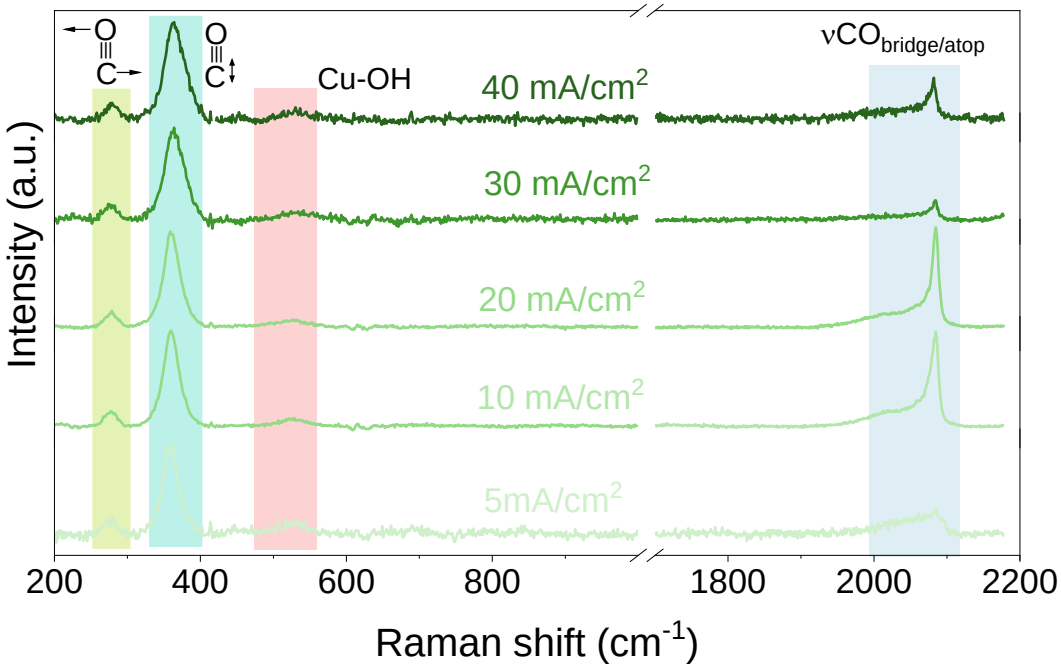


Supplementary Figure 4. Effect of cation concentration on the electric field at the cathode surface. This modeling result shows a small amount of cation can increase the cathode surface electric field and facilitate CO_2R.

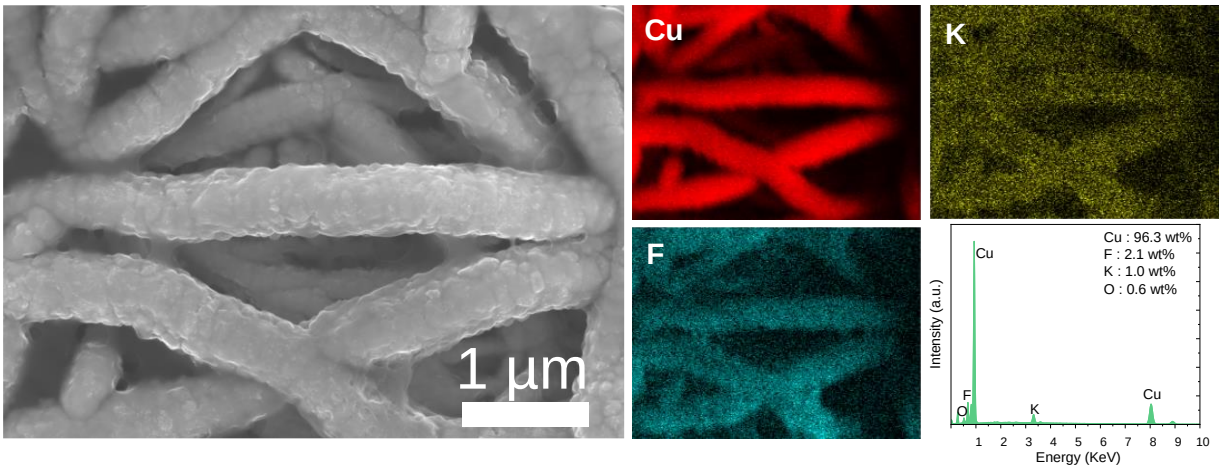
Supplementary Note 2 | Discussion on Surface-Enhanced Raman Spectroscopy – cation effect. To investigate the influence of alkali cation on acidic CO_2R efficiency, we incorporated 5 mM K_2SO_4 to the electrolyte of Raman cell. The CO intermediate is typically detected using surface-enhanced Raman scattering (SERS) in three distinct Raman shift regions: 280-290 cm^{-1} for the restricted rotation of adsorbed CO, 360-370 cm^{-1} for Cu-CO stretching, and 1800-2100 cm^{-1} for C-O stretching. The Raman shift within the 520-530 cm^{-1} range can be attributed to the Cu-OH band. It has been shown that a stronger intensity of the Cu-CO stretching band corresponds to a higher coverage of the CO intermediate and increased C_{2+} selectivity. Incorporating an alkali cation into the electrolyte results in a noticeable enhancement in the intensity of the restricted rotation of adsorbed CO and Cu-CO stretching, indicating a greater CO coverage. This promotes C-C coupling and C_{2+} formation (**Supplementary Figure 5 and 6**). The results indicate that the introduction of small amount of alkali cation close to the catalysts surface is critical to facilitate CO_2R. This result is an agreement with previous experimental observations in other configurations.[29–31] For example, Yang et al. observed an increase in CO FE as the salt concentration increased in a reverse BPM MEA with a silver-based cathode.[29] However, this trend may coincide with salt formation. Although this approach succors CO_2RR and suppresses HER compared to a salt-free electrolyte, the salt formation on the cathode limits the stable performance of the electrolyzer. Thus, we pursued strategies to enrich the catalyst surface with alkali cations while using a salt-free acidic anolyte.



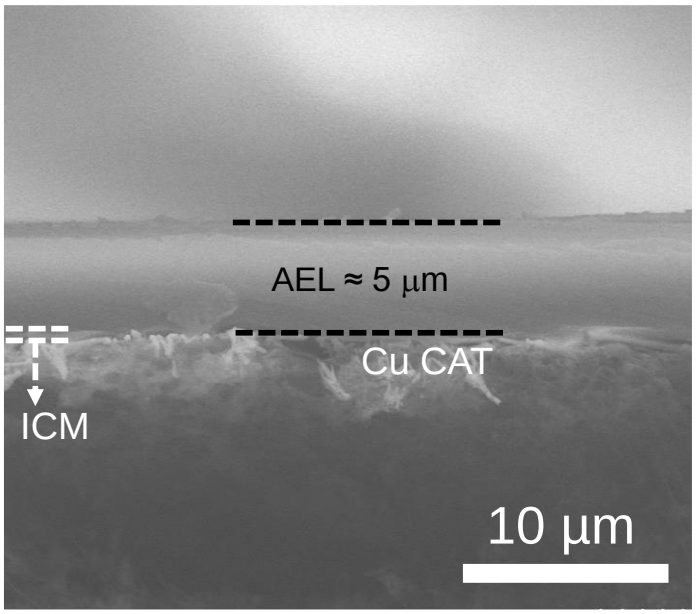
Supplementary Figure 5. *Operando* Raman analysis of Cu catalyst in acidic electrolyte. Assignment of the Operando Raman spectroscopy features of Cu electrode at different current densities in contact with 0.01 M H_2SO_4: adsorbed CO (*CO, 280-290 cm^{-1}) frustrated rotation, stretching of Cu-CO (360-370 cm^{-1}), Cu hydroxides (Cu-OH, 520-530 cm^{-1}), bridge-bound CO and atop-bound CO on Cu (Cu-CO_{bridge} and Cu-CO_{atop}, 1800-2100 cm^{-1}).



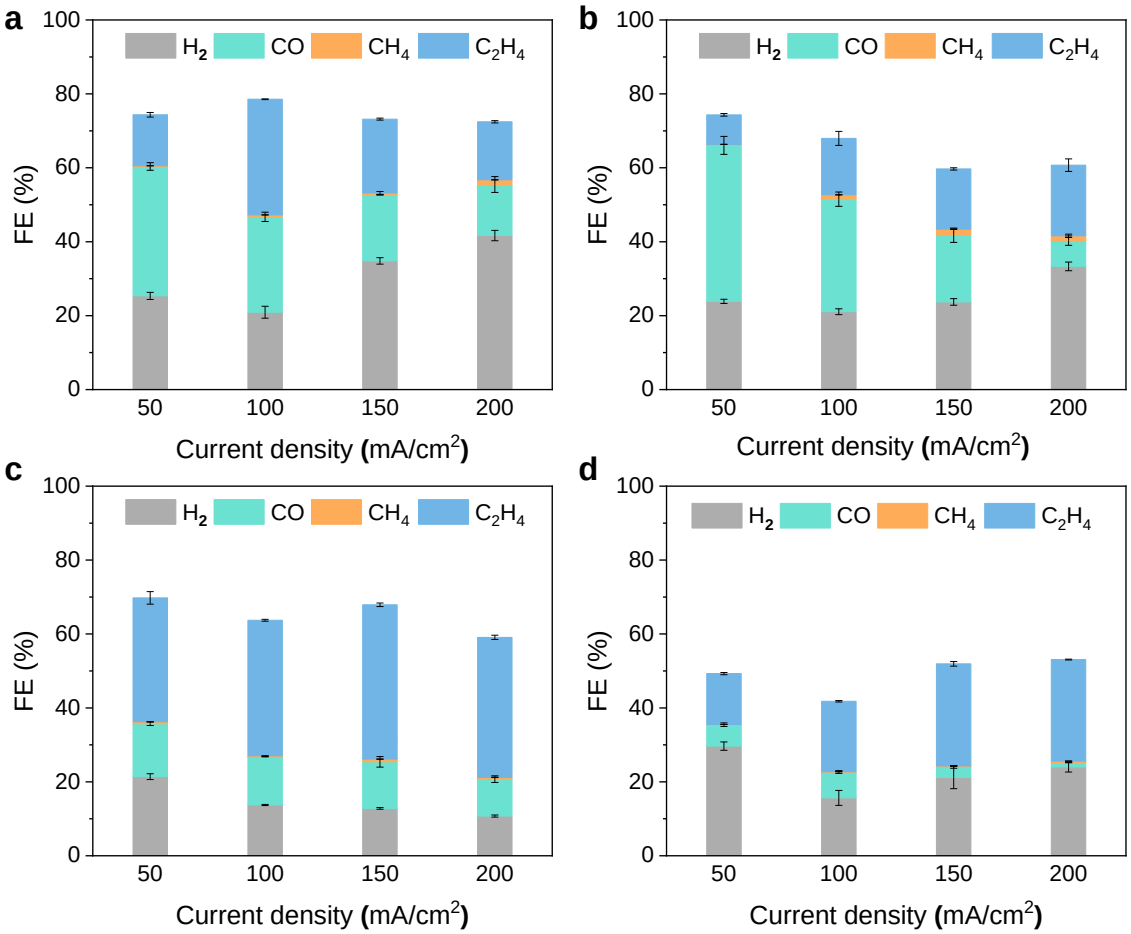
Supplementary Figure 6. *Operando* Raman analysis – effect of cation. Assignment of the Raman features (operando spectrum of Cu at different current densities) in contact with 0.01 M H_2SO_4 + 5 mM K_2SO_4. The results show that adding cation to the electrolyte increases the CO coverage that can facilitate CO_2R.



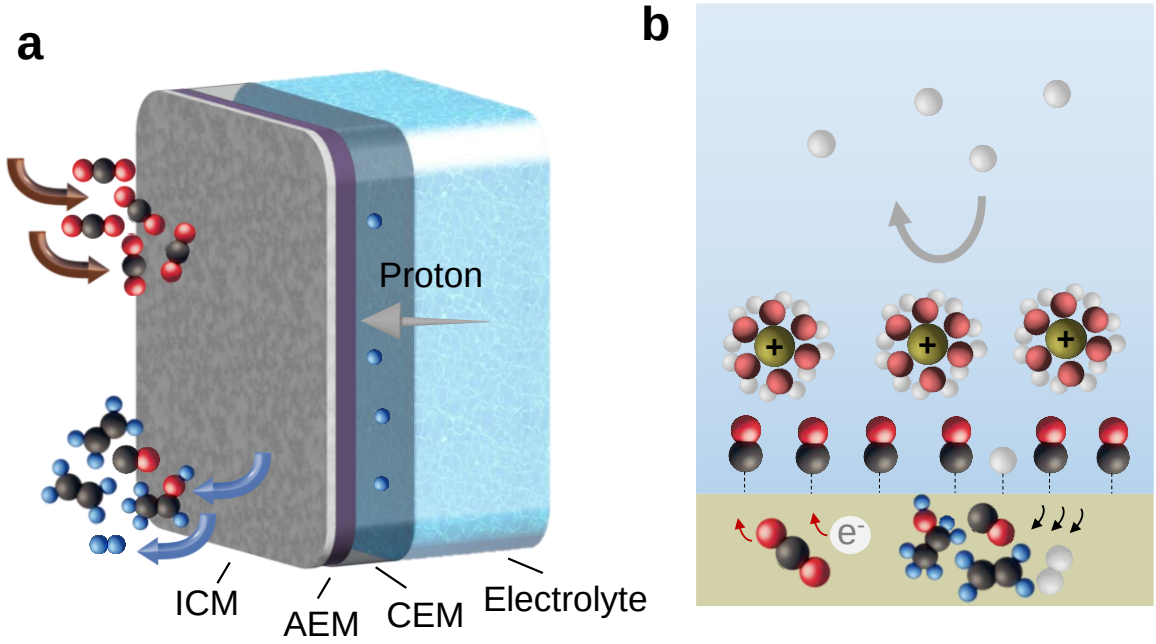
Supplementary Figure 7. SEM characterization of Cu catalyst coated with ICM. SEM images of Cu/PTFE coated with ICM with the corresponding Cu, F, and K EDS elemental mapping showing a homogeneous distribution of potassium over the catalyst surface.



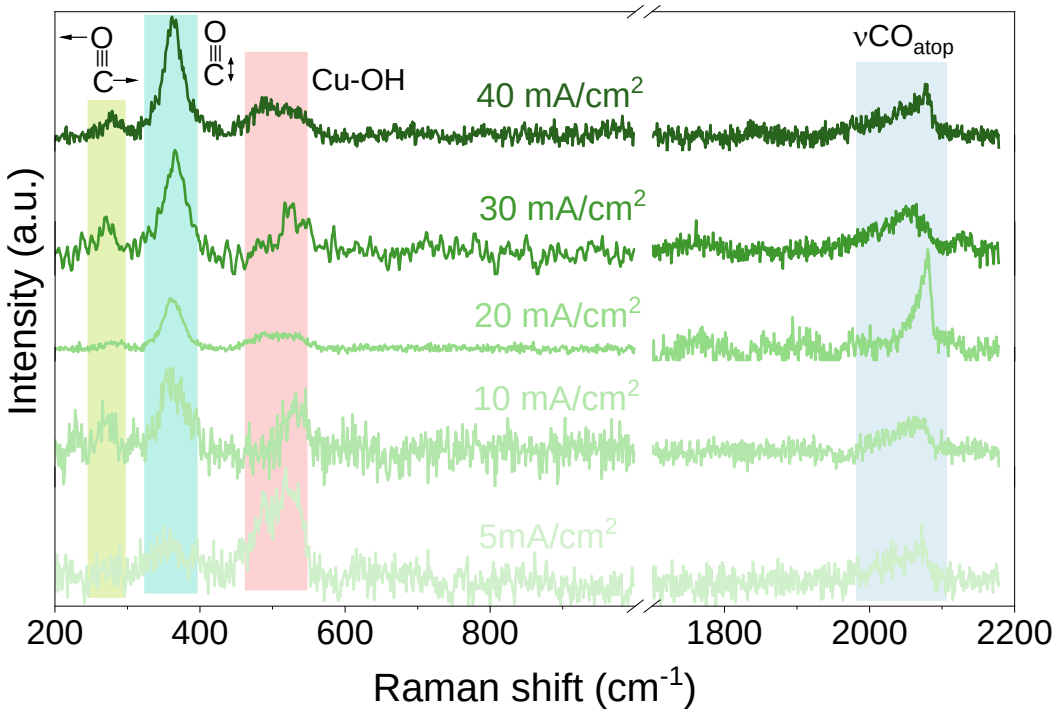
Supplementary Figure 8. SEM of anion exchange layer. Cross-sectional scanning electron microscopy image of gas diffusion electrode equipped with ICM and AEL. The AEL thickness is about 5 μm.



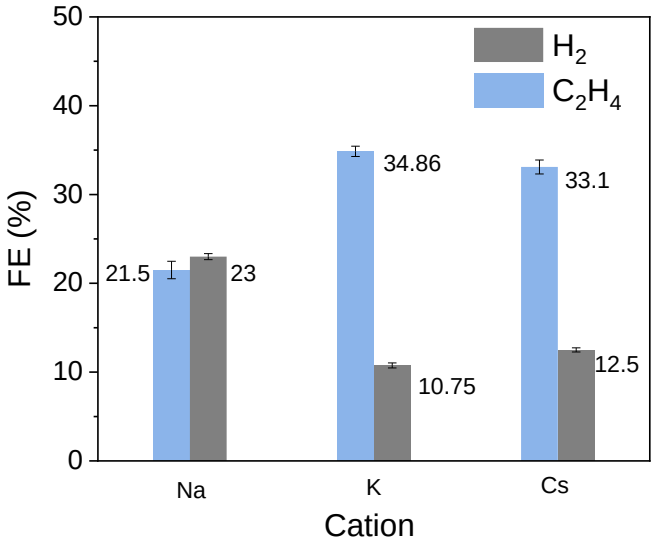
Supplementary Figure 9. Effect of ICM loading on CO_2R performance. Gas product FE for a Cu catalyst with different loading cation in the ICM layer: (a) 0, (b) 2, (c) 4, and (d) 8 µl/cm^2. The AEL loading was similar (3 mg/cm^2) for these experiments. The ratio of cation (K^+) to PFSA loading was 2. The case where there is zero cation loading indicates the absence of ICM. The plotted values are means of three replicates, while error bars indicate standard deviation.



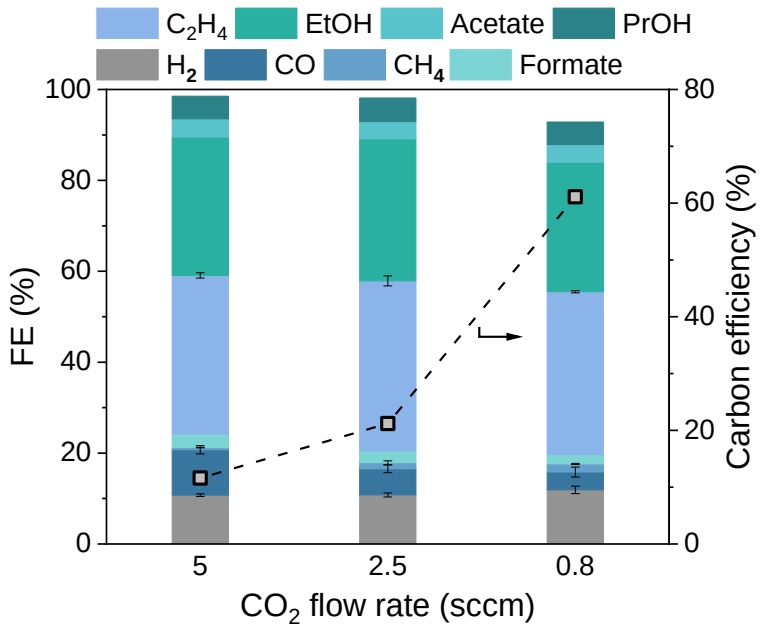
Supplementary Figure 10. Schematic of electrolyzer configuration and EDL for f-BPM MEA electrolyzer equipped with ICM. (a and b) ICM and AEL (AEM) ad-layers on the MEA electrolyzer. The ICM strategy can provide cation on the cathode surface and increase the local pH, improving CO_2 electroreduction.



Supplementary Figure 11. *Operando* Raman analysis of Cu catalyst equipped with ICM. Operando Raman spectroscopy studies of Cu catalyst equipped with ICM that indicates a higher intensity for the CO coverage peaks. Higher CO coverage facilitates C—C coupling and C_{2+} production.



Supplementary Figure 12. Effect of alkali cation type in the ICM on the CO_2R performance. Hydrogen and ethylene FE of ICM-modified PTFE–Cu electrodes with different cations inside the ICM layer at a concentration of 4 μl/cm^2. All experiments were performed at 200 mA/cm^2. The plotted values are means of three replicates, while error bars indicate standard deviation.

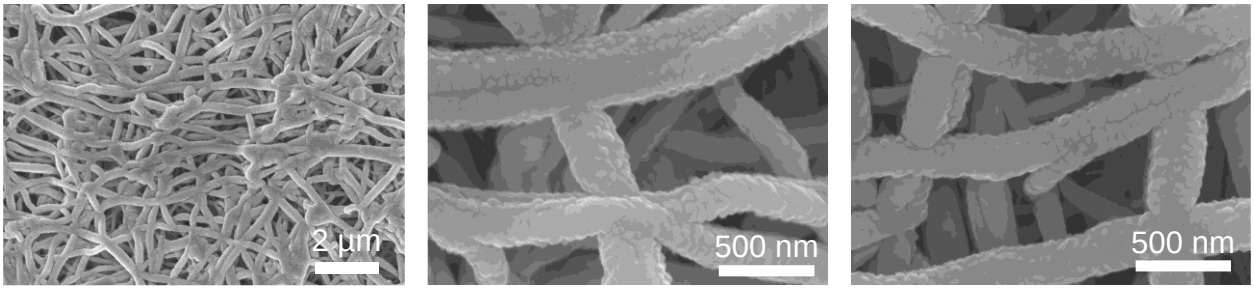


Supplementary Figure 13. Carbon efficiency of Cu catalyst equipped with ICM. CO_2R product distribution and carbon efficiency (dash line) of CO_2 to all products on an ICM-modified PTFE–Cu electrode at various CO_2 flow rates at 200 mA/cm^2. The plotted values for gas products are means of three replicates, while error bars indicate standard deviation.

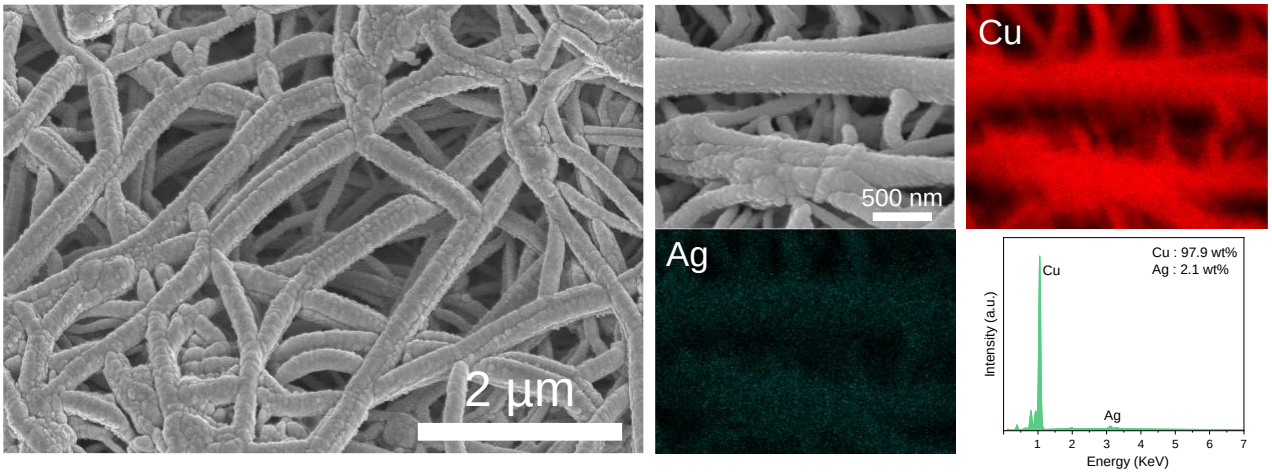
Supplementary Note 3 | Discussion on energy efficiency/full-cell voltage reduction. To allow CO_2 electrolyzers to be closer to industrial valorization, the electrolyzer energy consumption must be decreased. The current largest contributor to the total electrolyzer energy cost is the energy spent as the electricity – or cell potential – necessary to run the electrolyzer at industrially relevant current densities.[32,33] Thus, we focused our attention on improving the electrolyzer energy efficiency at an industrially relevant current density (200 mA/cm^2). We employed electrochemical Impedance Spectroscopy (EIS) to measure the IR drop across the MEA electrolyzer and identify the main sources of ohmic losses. Previous acid-fed CO_2 electrolyzers employed Nafion 117 as the CEM.[27,2834,35] We began our analysis performing CO_2R in MEA electrolyzer equipped with Nafion 117. By performing this analysis, through the Nyquist plots, we observed that ohmic losses are significantly high and increase a lot as we increase the current density (**Supplementary Table 8**). Two main sources of ohmic losses are membrane and contact resistance of the membrane/catalyst.

We then targeted ohmic overpotential reduction by using thinner CEMs. We tested three different types of CEM membrane: Nafion117, Nafion 212, and Nafion XL. Using a thinner membrane reduced the full-cell voltage with a larger reduction at higher current density; the applied cell potential decreased from 4.6 V to 4.0 V at 200 mA/cm^2 when Nafion 117 (183 µm) and Nafion XL (27 µm) were used, respectively (**Supplementary Figure 28**). We sought to further decrease the applied cell potential by pursuing the ohmic resistance caused by the contact between the anode and the membrane (**Supplementary Figure 28**). To decrease this resistance, we prepared a catalyst coated membrane (CCM) by spray-coating anode material (IrO_2 nanoparticles) on the CEM.[36] By using a thinner membrane such as Nafion XL-CCM instead of Nafion 117, a savings of 1.1 V in the full-cell voltage at 200 mA/cm^2 can be achieved which is of great

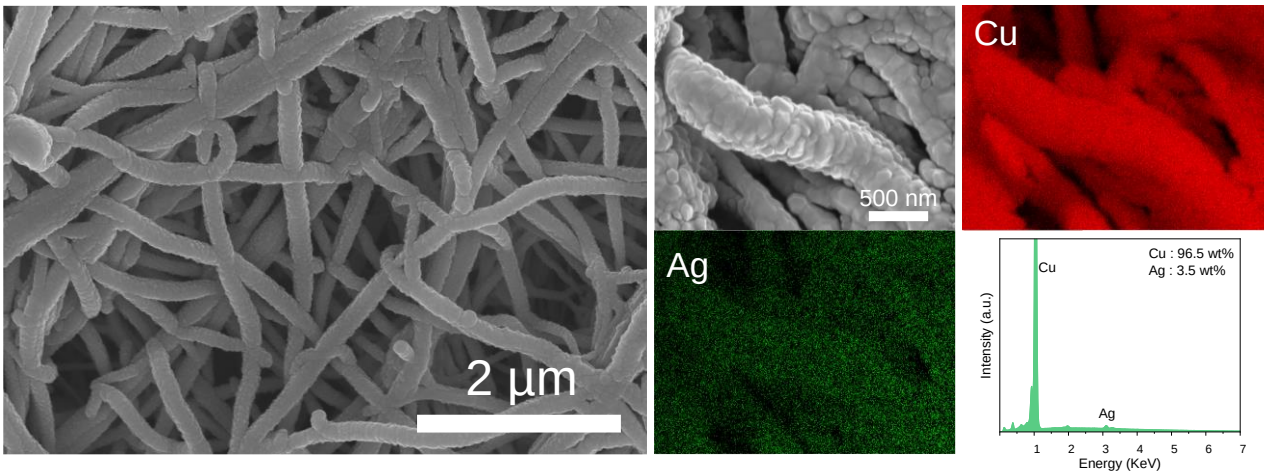
importance for commercial CO_2R viability. The CCM approach was utilized for the remainder of this work.



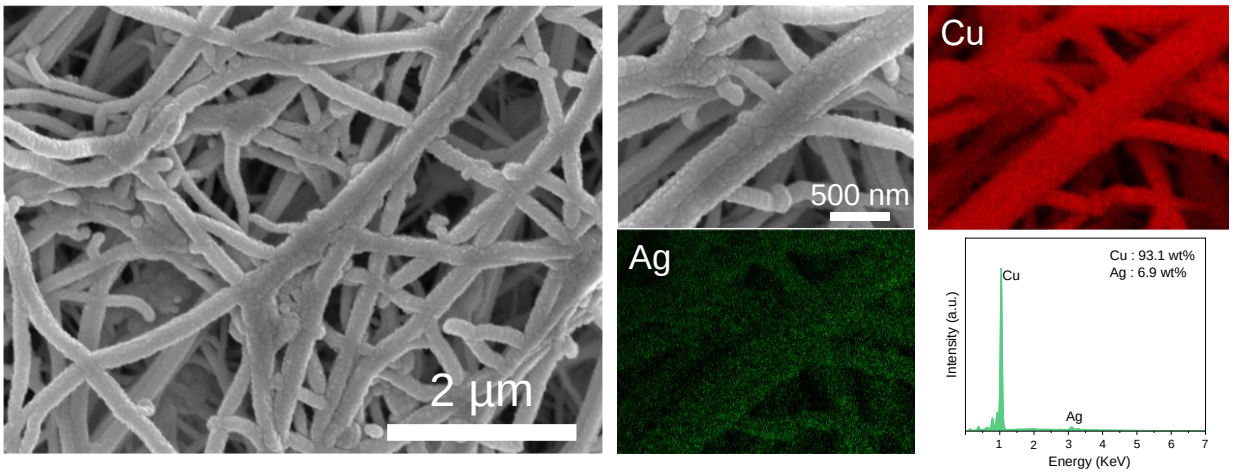
Supplementary Figure 14. SEM characterization of Cu catalyst. SEM images of Cu/PTFE at various magnifications.



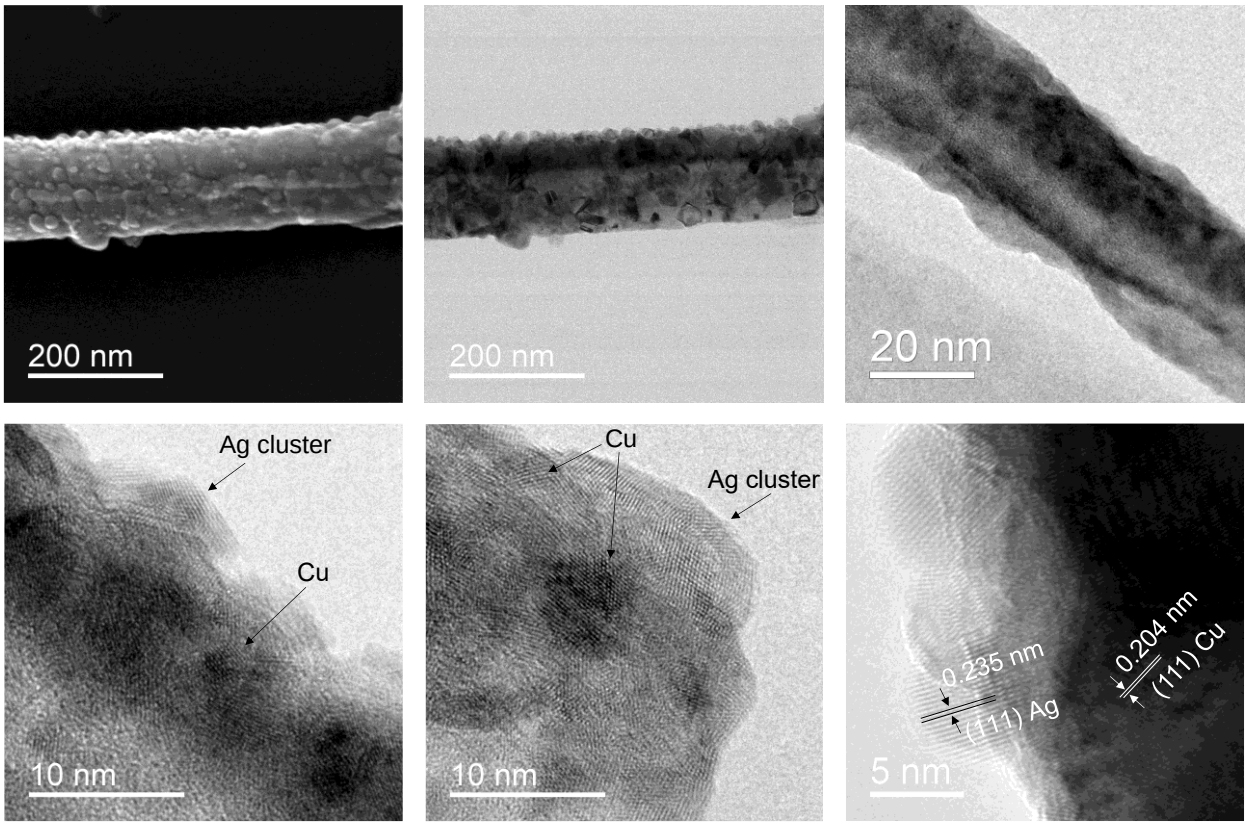
Supplementary Figure 15. SEM characterization of Cu-2 nm Ag catalyst. SEM images of Cu-2 nm Ag/PTFE at various magnifications with the corresponding Cu, and Ag elemental mapping, demonstrate a homogeneous distribution of Ag across the Cu surface. The nominal loading of Ag was 2 nm via sputtering.



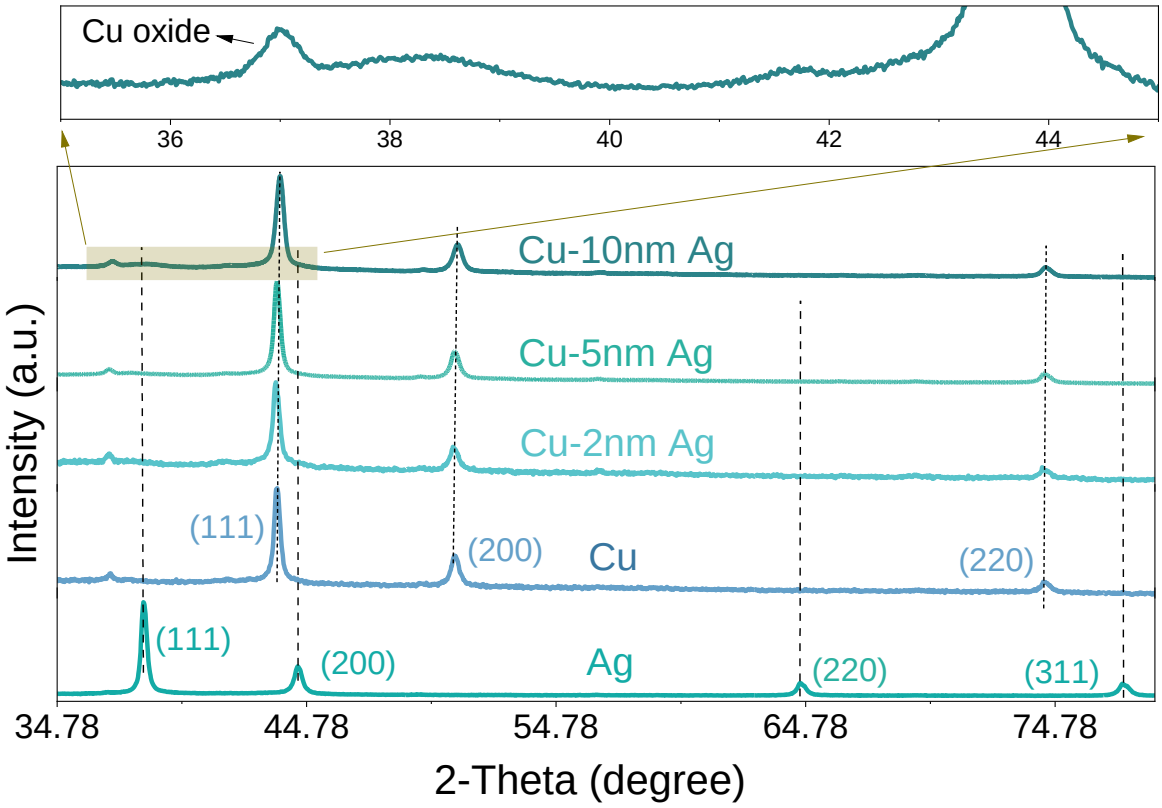
Supplementary Figure 16. SEM characterization of Cu-5 nm Ag catalyst. SEM images of Cu-5 nm Ag/PTFE at various magnifications with the corresponding Cu, and Ag elemental mapping, demonstrate a homogeneous distribution of Ag across the Cu surface. The nominal loading of Ag was 5 nm via sputtering.



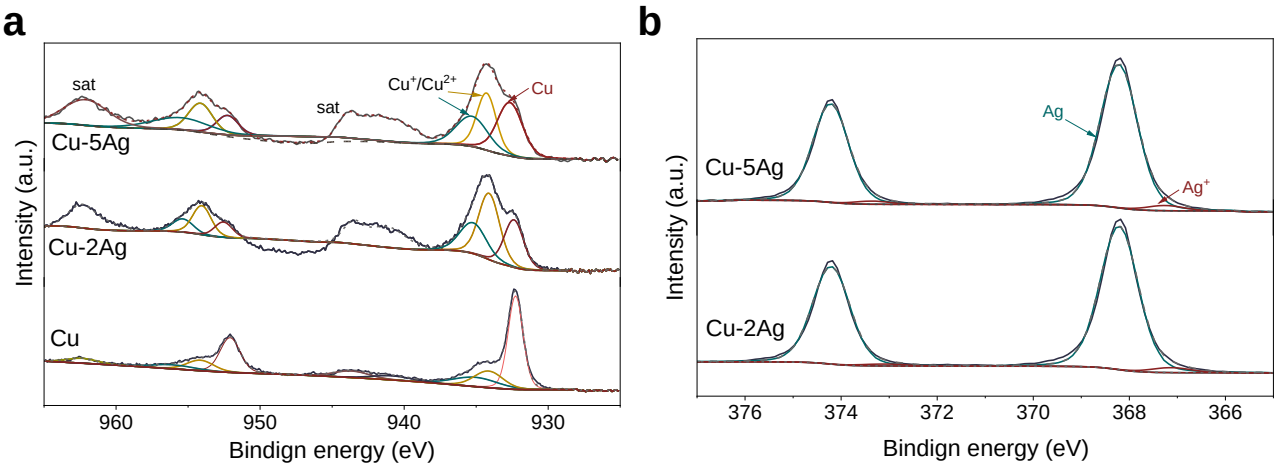
Supplementary Figure 17. SEM characterization of Cu-10 nm Ag catalyst. SEM images of Cu-10 nm Ag/PTFE at various magnifications with the corresponding Cu, and Ag elemental mapping, demonstrate a homogeneous distribution of Ag across the Cu surface. The nominal loading of Ag was 10 nm via sputtering.



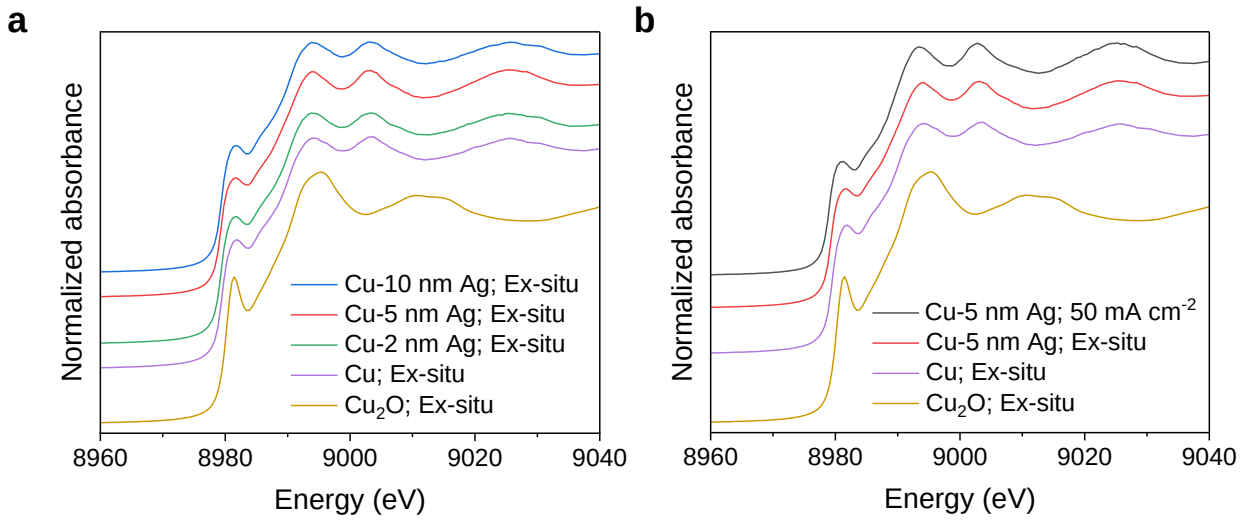
Supplementary Figure 18. STEM and HRTEM images of Cu- 5 nm Ag catalyst at different magnifications. HRTEM images indicate that the deposition of the Ag layer results in a nanocluster structure rather than a compact layer of Ag on the Cu surface. This could be attributed to the porous nature of the Cu-PFTE (SEM images on Supplementary Figure S13-S16) and the limited thickness of the Ag layer (5 nm), which prevented complete coverage of the Cu surface.



Supplementary Figure 19. XRD patterns of catalysts. X-ray diffraction patterns of PTFE-Ag, PTFE-Cu, and PTFE-Cu-Ag electrodes with different loadings of silver (2 nm, 5 nm, and 10 nm). No clear silver peak was observed in Cu-2 nm Ag and Cu-5 nm Ag catalysts, which can be attributed to the small Ag cluster size; however, XRD reveals a subtle, less crystalline Ag peak in the Cu-10 nm Ag catalyst.

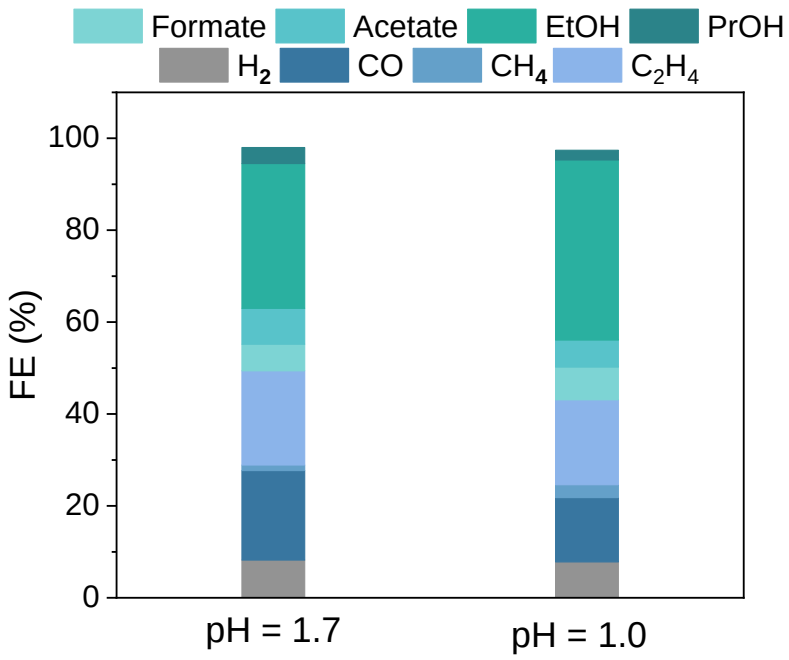


Supplementary Figure 20. High-resolution XPS spectra. (a) Cu2p and (b) Ag3d. The XPS analyses clearly shows the presence of Ag on the surface of Cu.



Supplementary Figure 21. Ex-situ and in-situ X-ray absorption spectroscopy (XAS) spectra. (a) Ex-situ XAS analyses for the Cu-Ag catalysts reveal that bulk Cu primarily exists in its metallic form (Cu^0). The ex-situ spectra were recorded in transmission mode. (b) In-situ and ex-situ XAS analyses for the Cu-5 nm Ag catalyst reveal that during CO_2R, the Cu primarily exists in its metallic form (Cu^0). The in-situ XAS spectra of the Cu K-edge were recorded in fluorescence mode using a custom in-situ XAS flow cell at a constant current of 50 mA cm^{-2}.

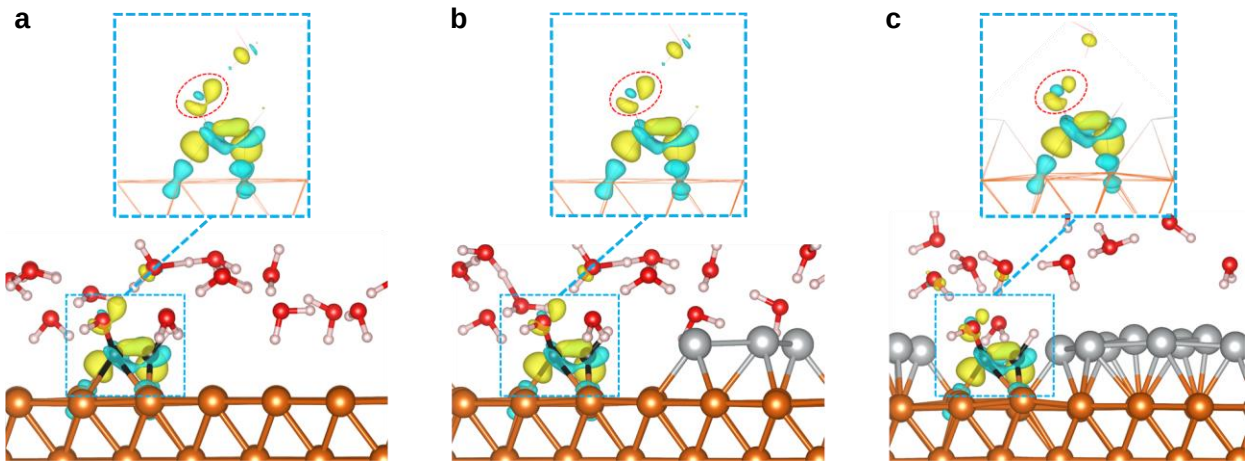
The ex-situ spectra show the copper to be predominantly in the metallic $Cu^{(0)}$ state. All samples retain their $Cu^{(0)}$ state under reaction conditions, as evidence by the unchanged spectra during in-situ characterization.



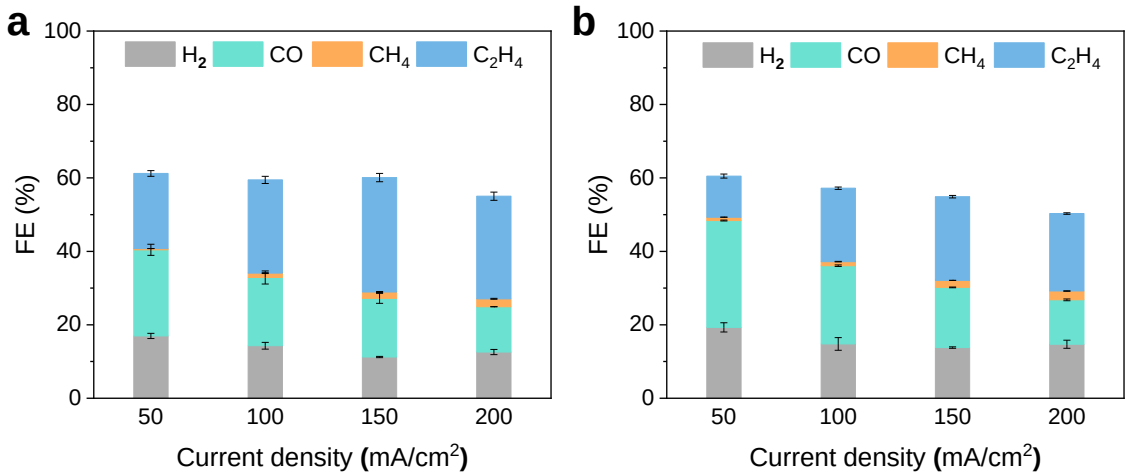
Supplementary Figure 22. Effect of pH on ethanol production. CO_2R product distribution for a Cu-5Ag catalysts at 300 mA/cm^2 with different electrolyte. The experiment was performed in a flow cell to study the effect of pH on ethanol and ethylene formation. The electrolytes were 0.01M H_2SO_4 + 3M KCl (pH = 1.7) and 0.05M H_2SO_4 + 3M KCl (pH = 1.0). This experiment shows that increasing the H^+ concentration (lower pH) facilitates ethanol formation in acidic CO_2R.

Supplementary Note 4 | DFT modelling of Cu-Ag interface for CO_2R. Previous studies have established Ag as a catalyst for producing CO from CO_2, while Cu facilitates C-C coupling, leading to formation of C_{2+} products. In our Cu-Ag system, Cu promotes the production of C_{2+} products by coupling the CO generated by Ag clusters on the Cu surface. Therefore, the Cu-Ag interface, rather than the Ag cluster itself, plays a critical role in determining the reaction selectivity toward C_{2+} products. To gain mechanistic insight into the effect of Ag on the formation of C_2 intermediates, we conducted DFT calculations to investigate the Cu-Ag interface on Cu surfaces. First, AIMD simulations were conducted to optimize the hexagonal charged water layer on Cu-Ag surface, comprising five water molecules and one hydronium (H_3O^+), to account for the field and solvation effects. Although alkali cations were not included in this model, it still provides a reliable framework for investigating the reaction energy profile of C_2 intermediate formation with reasonable computational accuracy and speed, as suggested by Montoya et al.[37]

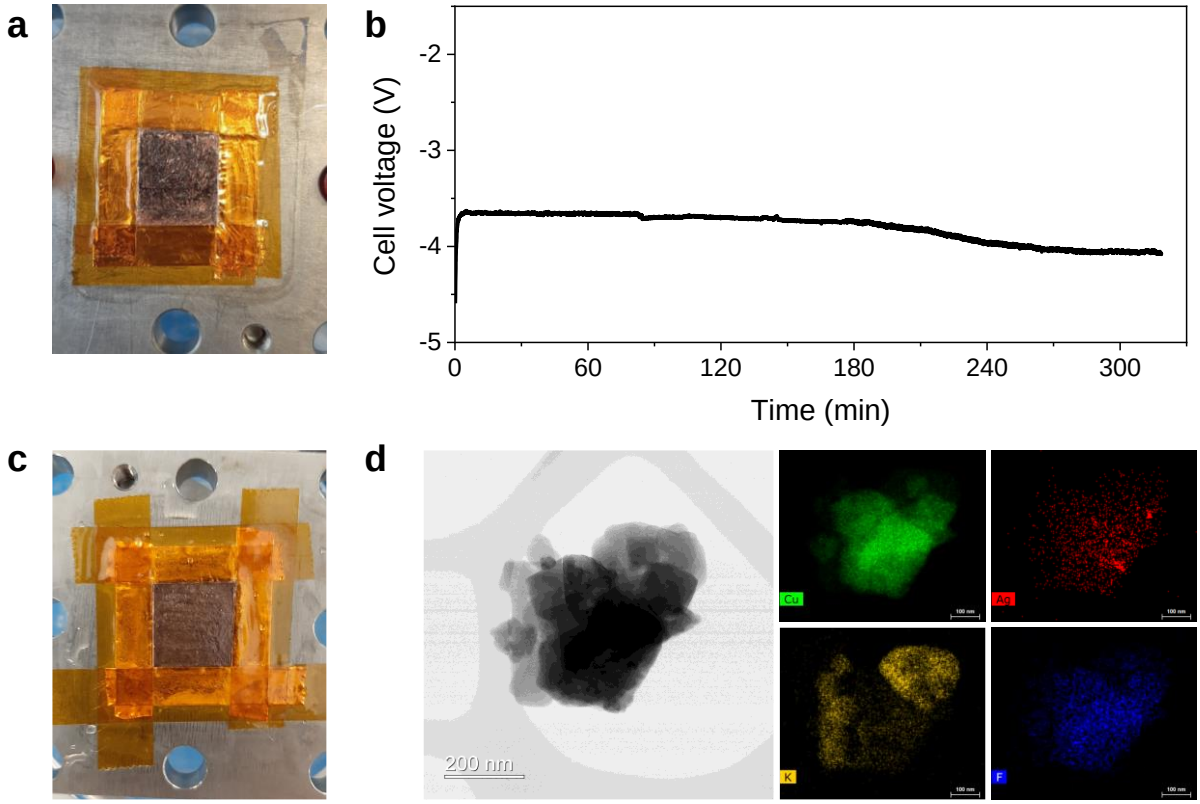
Next, DFT calculations were applied to study the key reaction intermediates involved in ethanol and ethylene formation, such as $HCOOH^*$, $HCCHO^*$, H_2CCO^* on the Cu-Ag surface optimized by AIMD. Our DFT calculations revealed that the Cu-Ag interface indeed favours the formation of C_2 intermediates on the path to ethanol production compared to Cu surface, providing valuable mechanistic insights into the Cu-Ag catalyst's superior ethanol production performance.



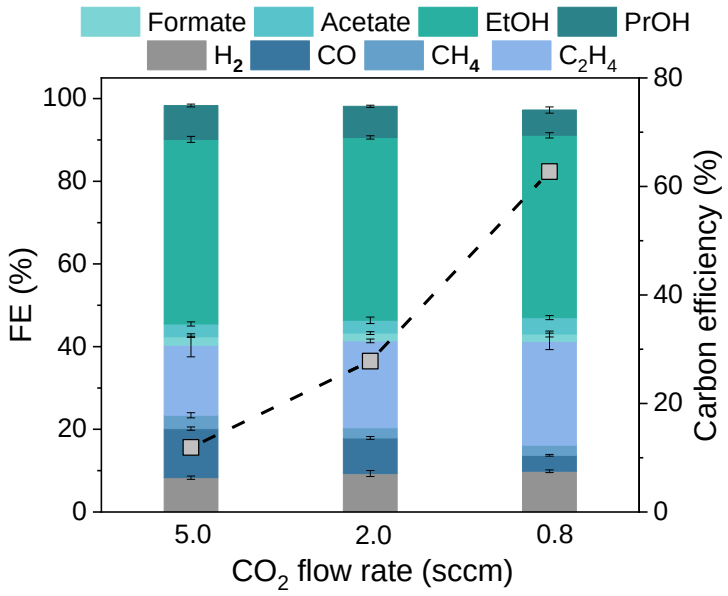
Supplementary Figure 23. Electron density difference plots for Cu and Cu-Ag catalysts. Surface configurations of (a) Bare Cu, (b) Cu with 3 atoms of Ag on top, and (c) Cu with 9 atoms of Ag on top (which create an Ag cluster through which C–O bond scission is thermodynamically unfavourable on the interface of Cu-Ag clusters). Ag clusters on Cu sites would increase the local concentration of CO. Thereby, a local CO reservoir is created. The availability of CO would promote ethanol production on the Cu sites by stabilizing the C–O bond and H* spillover mechanism.



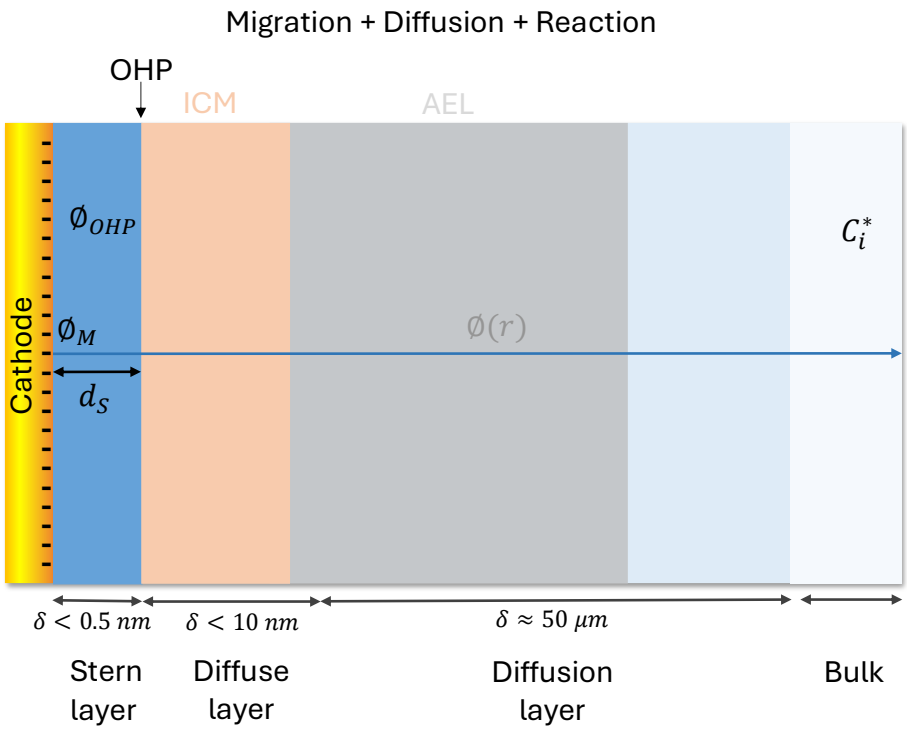
Supplementary Figure 24. Effect of deposition of AgNP on CO_2R performance. CO_2R product distribution for a PTFE/Cu catalyst with Ag nanoparticles (AgNPs, 20 nm) coating, corresponding to sputtering loadings of 5 nm and 10 nm. The experiments demonstrate that adding AgNPs to the Cu catalyst increases CO FE, which correlates with AgNP acting as a CO catalyst. In this work, sputtering was employed for Ag deposition to access very small Ag nanoclusters (1-10 nm loading), thereby increasing the Cu-Ag interfaces and ethanol production. The plotted values are means of three replicates, while error bars indicate standard deviation.



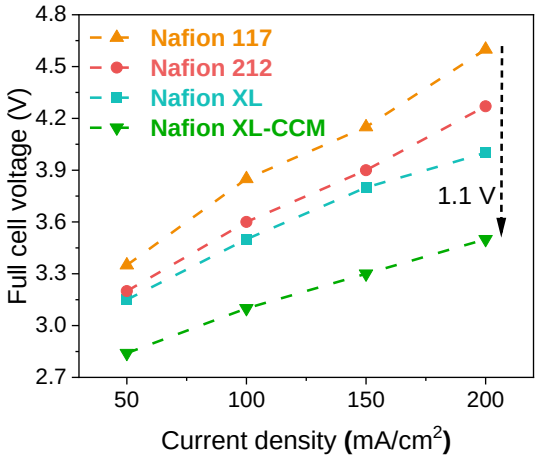
Supplementary Figure 25. Stability of Cu-5Ag catalyst with and without the ICM adlayer. (a) Image of post-electrolysis catalyst surface without the ICM adlayer. (b) Stability of the Cu-Ag without the ICM adlayer. It shows that the ICM layer enhances the stability of the catalyst (Figure 4). (c) Image of post-electrolysis catalyst surface with the ICM adlayer after the stability test. (d) STEM image and corresponding EDS mapping of a Cu-Ag catalyst equipped with the ICM adlayer after stability testing. The images and mapping confirm the presence of ICM (K and F) after CO_2 electrolysis.



Supplementary Figure 26. Carbon efficiency of a Cu-Ag catalyst. The carbon efficiency of a Cu-Ag (with a loading of 5 nm) catalyst equipped with ICM as a function of CO_2 flow rate at 200 mA/cm^2. The plotted values are means of three replicates, while error bars indicate standard deviation.



Supplementary Figure 27. **Multiphysics simulation domain.** 1D domain for the simulation of space charge density, E-field, and pH.



Supplementary Figure 28. Full-cell voltage for acid-fed CO_2 MEA electrolyzer with different CEM membranes. Using a thinner membrane with a catalyst coating (CCM; Nafion XL-CCM *vs* Nafion 117) could save 1.1 V in the full-cell voltage at 200 mA cm^{-2}.

Supplementary Table 1. Total energy cost comparison for ethanol production. Comparison of total energy cost for different CO_2 to ethanol electrolyzers. The comparison is limited to electrolyzers working at total current densities ≥ 100 mA/cm^2

Parameters	**Alkaline CO_2R**			**Neutral CO_2R**				**Acidic CO_2R**						
Cell voltage (V)	2.8	3	3	3.9	3.7	3.5	3.7	3.8	3.75	3.5	4.2	3.8	3.7	3.5
EtOH FE (%)	26	40	69	23	41	50	52	20	21	12	12	9	16	44
Current Density (mA/cm^2)	100	900	271	300	600	190	125	200	100	500	1200	100	200	200
Carbon efficiency (%)	4	5*	5*	15*	15*	7	5	50	40	70	70	85	75	63
EtOH conc. (wt.%)	1	1	1	4	1	1	1	13	4	1	1	1	1	1
EtOH crossover (%)	60%	60%	60%	50%	60%	80%	80%	60%	20%	60%	60%	60%	60%	60%
Electrolyzer specific energy distribution (GJ (ton product)$^{-1}$)														
CO_2 Capture	45	29	23	62	37	23	23	74	60	105	105	93	92	28.2
Electrolyzer electricity	271	188	110	426	227	176	179	478	449	733	880	1061	589	199.9
Ethanol distillation	22	22	22	16	17	22	22	15	22	22	22	22	22	22
Cathode gas separation	578	285	235	118	75	146	220	45	55	28	28	10	19	10.2
Anode gas separation	88	57	33	100	56	46	44	-	-	-	-	-	-	-
Total energy cost (GJ/ton EtOH)	**1004**	**581**	**423**	**723**	**412**	**413**	**487**	**612**	**586**	**887**	**1034**	**1187**	**721**	**260.3**
Ref.	38	39	40	41	42	43	44	45	28	26	30	46	31	**This work**

*These are not experimental values; the maximum theoretical carbon efficiency was considered for the energy cost calculation.

Supplementary Table 2. Ethanol FE in the recent literature. Comparison of ethanol FE for different CO_2 to ethanol electrolyzers. The comparison is limited to electrolyzers working at total current densities ≥ 100 mA/cm^2 and EtOH FE > 50%

System	Electrolyte	Catalyst	Ethanol FE (%)	Carbon efficiency (%)	E (FC / vs RHE)	J (mA cm^{-2})	Ref.
Flow cell	3M KOH	Im-Mo_3P	77	... < 5*	-0.5 vs RHE	90-120	47
	1M KOH	V-doped Cu_2Se	68	... < 5	-0.9 vs RHE	300	48
	1M KOH	Cu-SC_{12}	57	... < 5	-1.2 vs RHE	~200	49
	1M KOH	Ag@$Cu_{0.05}$	53	... < 5	-1.0 vs RHE	320	50
	1M KOH	K–F–Cu	53	... < 5	-0.7 vs RHE	432	51
MEA	1M KOH	Ag^+-Doped InSe	69	... < 5	3	271	40
	0.02 M H_2SO_4	**Cu-Ag**	**45**	**63**	**3.5**	**200**	**This work**

* 5% is the maximum carbon efficiency for the alkaline electrolyzer.

Supplementary Table 3. Diffusion-coefficients in $m^2 s^{-1}$.[2]

Constant	Value
D_{H^+}	9.311×10^{-9}
D_{Na^+}	1.334×10^{-9}
D_{K^+}	1.957×10^{-9}
D_{Cs^+}	2.06×10^{-9}
D_{OH^-}	5.273×10^{-9}
D_{CO_2}	1.91×10^{-9}
$D_{HCO_3^-}$	1.185×10^{-9}
$D_{CO_3^{2-}}$	0.923×10^{-9}

Supplementary Table 4. Solvation sizes in $m.^2$

Constant	Value
a_{H^+}	0.56×10^{-9}
a_{Na^+}	0.716×10^{-9}
a_{K^+}	0.662×10^{-9}
a_{Cs^+}	0.658×10^{-9}
a_{OH^-}	0.6×10^{-9}
a_{CO_2}	0.23×10^{-9}
$a_{HCO_3^-}$	0.8×10^{-9}
$a_{CO_3^{2-}}$	0.788×10^{-9}

Supplementary Table 5. **Rate constants used for modelling**. [52]

Constant	Value	Units
k_{1f}	3.71×10^{-2}	s^{-1}
k_{1r}	1.99×10^{4}	s^{-1}
k_{2f}	59.99	s^{-1}
k_{2r}	2.28×10^{7}	$mol^{-1}m^{3}s^{-1}$
k_{3f}	2.23	$mol^{-1}m^{3}s^{-1}$
k_{3r}	5.23×10^{5}	s^{-1}
k_{4f}	6.0×10^{6}	$mol^{-1}m^{3}s^{-1}$
k_{4r}	1.07×10^{6}	s^{-1}
k_{wf}	2.4×10^{-2}	$mol\ m^{3}s^{-1}$
k_{wr}	2.4×10^{6}	$mol^{-1}m^{3}s^{-1}$

Supplementary Table 6. **h parameters used to estimate Sechenov's constant** in $m^3 kmol^{-1}$.[53]

Constant	Value
h_{Na^+}	0.1143
h_{K^+}	0.0922
h_{Cs^+}	0.0759
h_{OH^-}	0.0839
$h_{HCO_3^-}$	0.0967
$h_{CO_3^{2-}}$	0.1423
$h_{SO_4^{2-}}$	0.1117
$h^0_{CO_2}$	-0.0172
$h^T_{CO_2}$	-0.00038

Supplementary Table 7. **Contributions to Gibbs free energies of gas species and adsorbates**, including zero-point energy (ZPE), enthalpic temperature correction ($\int C_P \mathrm{d}T$), entropy (S). H_2 refers to the energy used in the computational hydrogen electrode model

Species	$\boldsymbol{ZPE}$ $(\boldsymbol{eV})$	$\int \boldsymbol{C_P}\mathbf{d}\boldsymbol{T}$ $(\boldsymbol{eV})$	$-\boldsymbol{TS}$ (**eV**)	$\boldsymbol{G}-\boldsymbol{E}$ $(\boldsymbol{eV})$
CO_2(g)	0.31	0.10	-0.66	-0.25
H_2O(l)	0.57	0.10	-0.67	0
H_2(g)	0.27	0.08	-0.40	-0.05
$HCCO^*$	0.58	0.11	-0.21	0.48
$HCCOH^*$	0.91	0.09	-0.15	0.85
HCCHO*	0.85	0.11	-0.25	0.71
H_2CCO*	0.88	0.11	-0.20	0.79
H^*	0.17	0.01	-0.01	0.17

Supplementary Table 8. Full cell voltage and RI for MEA electrolyzer at different current densities. Nafion 117 was used as the CEM with 0.01 M H_2SO_4 as anolyte

Current density (mA/cm^2)	50	100	200
Full cell voltage (v)	3.5	3.8	4.6
RI (v)	0.26	0.52	0.93

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